

SUBJECT- MATH CLASS-8

Q1. Fill in the blanks: The _____ is the middle value of a data set when the values are arranged in ascending or descending order.

Q2. If every value in a data set is increased by 5, the new mean will increase by exactly _____.

Q3. MCQ: Which of the following is most suitable for showing how a city's temperature changes over the twelve months of a year?

- (a) Pictograph
- (b) Line Graph
- (c) Dot Plot
- (d) Pie Chart

Q4. True or False: The sum of the distances of all data points to the left of the mean is always equal to the sum of the distances of all points to the right of the mean.

Q5. In a spreadsheet, if you want to find the average of marks located in cells C2, C3, C4, and C5, which formula should you use?

Q6. The mean of five numbers is 12. If a new number, 18, is added to the collection, calculate the new mean.

Q7. Consider the data set: 10, 12, 12, 15, 20, 25, 30. If the value '30' is replaced by '10', how will the median change?

Q8. A student scores 45, 47, and 43 in three subjects. What must they score in the fourth subject to have a mean score of 46?

Q9. Briefly explain the "Balancing Act" of the mean using the concept of distances on a number line.

Q10. Look at the table below and find the mean frequency:

VALUE	11	12	10
FREQUENCY	5	6	4

Q11. Case Study: A class of 20 students had an average height of 155 cm. It was later discovered that the measuring tape was faulty and under-reported every student's height by 2 cm. What is the actual correct mean height? Prove your answer using algebra.

Q12. Reasoning: Suppose you have a data set where the mean is 25. If you add two new values, 20 and 30, to this data set, will the mean change? Justify your answer using the distance principle.

Q13. Word Problem: A shopkeeper records the number of umbrellas sold over a week: 5, 2, 0, 15, 25, 30, 10. (a) Find the median sales. (b) If the shopkeeper wants to show the "trend" of sales to prepare for next year's monsoon, which graph should he use and why?

Q14. In a frequency distribution table of 40 observations, the median is the average of which two positions? If the first 18 positions are the value '7' and the next 5 positions are the value '9', what is the median?

Q15. If the mean of x_1, x_2, x_3, x_4 is A, prove algebraically that the mean of $3x_1, 3x_2, 3x_3, 3x_4$ is 3A.